AS · Unit 2 · WPH12 Waves & Electricity

Particle Nature of Light

Is light a wave or a stream of particles? Physicists argued about it for two hundred years, and the answer turned out to be "both, depending on the experiment". On this page you'll see why the photoelectric effect forced us to think of light as photons, how atoms emit and absorb light at exact energies, and how electrons, which we think of as particles, can diffract like waves.

SPEC 53 Electron diffraction: evidence that electrons behave as waves

WHAT THIS SPEC POINT ASKS

Understand how electron diffraction experiments give evidence that electrons have a wave nature.

The experiment

1. In an evacuated **electron diffraction tube**, electrons from a heated cathode are **accelerated** through a p.d. of a few kV.
2. The beam passes through a very thin film of **polycrystalline graphite**.
3. On the fluorescent screen you see **concentric rings**, not a single spot.

Why it is evidence of wave behaviour

- Rings like these are a **diffraction / interference pattern**: the same thing light does through a grating. Particles alone would give one spot (or a blur).
- The regular layers of carbon atoms act like a **grating**. Diffraction is strong because the electrons' **de Broglie wavelength** (54) is **similar to the spacing between the atoms** (about 10^{-10} m).
- **Raise the accelerating p.d.** → electrons go faster → greater momentum → **shorter wavelength** ($\lambda = h/p$) → less diffraction → **rings get smaller**.
- The same electrons hit the screen at single points, like particles. They show both behaviours.

WORKED EXAMPLE

why an ordinary grating won't do

Electrons accelerated through 2500 V have $\lambda = 2.46 \times 10^{-11}$ m (see 54). A light grating with 500 lines per mm has $d = 2.00 \times 10^{-6}$ m.

$\sin\theta = \lambda/d = 2.46 \times 10^{-11} \div 2.00 \times 10^{-6} = 1.2 \times 10^{-5} \rightarrow \theta \approx 7 \times 10^{-4}°$.

Far too small to see. The gap must be comparable to λ, which is why atomic spacings are used.

WORKED EXAMPLE · MULTI-STEP

In an electron diffraction tube, electrons are accelerated through 3.5 kV. (a) Show that their de Broglie wavelength is about 2×10^{-11} m. (b) The p.d. is increased to 5.0 kV. Explain what happens to the rings on the screen. (c) Calculate the speed a neutron (mass 1.67×10^{-27} kg) would need to have the same wavelength as in (a).

Show solution

(a) $\lambda = h/\sqrt{2meV}$:

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.11 \times 10^{-31} \times 1.60 \times 10^{-19} \times 3500}} = \frac{6.63 \times 10^{-34}}{3.19 \times 10^{-23}} = 2.08 \times 10^{-11}\ \text{m} \approx 2 \times 10^{-11}\ \text{m}$$

(b) The electrons gain more kinetic energy and momentum, so $\lambda = h/p$ decreases (to $2.08 \times 10^{-11} \times \sqrt{3.5/5.0} = 1.74 \times 10^{-11}$ m). Shorter waves diffract through smaller angles, so the rings get **smaller** (and closer to the central spot).

(c) Same wavelength means the same momentum, so $v = h/(m\lambda)$:

$$v = \frac{6.63 \times 10^{-34}}{1.67 \times 10^{-27} \times 2.08 \times 10^{-11}} = 1.9 \times 10^{4}\ \text{m s}^{-1}$$

The neutron is about 1800 times heavier than the electron, so it needs to go about 1800 times slower to have the same momentum.

MARK-SCHEME WORDING

- "Electrons **diffract** as they pass through the graphite", so "electrons **can behave as / show the properties of** waves".
- "The (de Broglie) wavelength of the electrons is **similar to the spacing** between the atoms"; graphite has a "regular/ordered structure" acting as a grating.
- Higher p.d.: "greater momentum, so λ = h/p is smaller, so the rings get smaller".

Often asked as: interpret ring patterns (2–4 marks), effect of changing the accelerating p.d., MCQs on what counts as evidence of wave behaviour, why a neutron or an ordinary grating wouldn't work.

COMMON MISTAKE

Writing "electrons **are** waves". They **behave as** waves here (WPH12 Oct 2024 examiner report).

Saying "wave behaviour" without naming **diffraction** or interference.

Thinking attraction between charges, or electrons hitting the screen as dots, is evidence of wave nature (it isn't).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Is a particle suitable for diffraction by atoms? Compare its wavelength with the atomic spacing. A neutron with λ = 2.8 × 10⁻¹⁴ m is far smaller than an atom (about 10⁻¹⁰ m), so it would not diffract noticeably. The final mark is the numerical comparison (WPH12 Oct 2020).

SPEC 54 The de Broglie equation

WHAT THIS SPEC POINT ASKS

Use λ = h/p for the wavelength of a moving particle.

The equation

de Broglie wavelength

FORMULA SHEET

$$\lambda = \frac{h}{p}$$

λ: wavelength / m $h = 6.63 \times 10^{-34}$ J s $p = mv$: momentum / kg m s⁻¹

λ = de Broglie wavelength (m), h = Planck constant = 6.63 × 10⁻³⁴ J s, p = momentum (kg m s⁻¹) = mass (kg) × velocity (m s⁻¹). On the formulae sheet.

- Every moving particle has a wavelength. Bigger momentum → shorter wavelength.
- Everyday objects have huge momentum compared with h, so their wavelengths are absurdly small and no wave behaviour is ever seen.
- For an electron accelerated from rest through p.d. V: kinetic energy gained = eV = ½mv², so find v, then p = mv, then λ.

WORKED EXAMPLE

electron and ball

(a) An electron (m = 9.11 × 10⁻³¹ kg) is accelerated from rest through 2500 V.

E_k = eV = 1.60 × 10⁻¹⁹ × 2500 = 4.00 × 10⁻¹⁶ J.

v = √(2E_k/m) = √(2 × 4.00 × 10⁻¹⁶ ÷ 9.11 × 10⁻³¹) = 2.96 × 10⁷ m s⁻¹.

p = mv = 9.11 × 10⁻³¹ × 2.96 × 10⁷ = 2.70 × 10⁻²³ kg m s⁻¹.

λ = h/p = 6.63 × 10⁻³⁴ ÷ 2.70 × 10⁻²³ = **2.46 × 10⁻¹¹ m** (3 s.f.). Similar in size to atomic spacings, so it diffracts.

(b) A 0.058 kg tennis ball at 50 m s⁻¹: λ = 6.63 × 10⁻³⁴ ÷ (0.058 × 50) = **2.3 × 10⁻³⁴ m**. Far too small to show any diffraction.

Wavelength of an electron accelerated through a p.d. V

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. The electron starts at rest. Moving through a p.d. V, it gains energy $W = QV = eV$, all as kinetic energy:

$$\tfrac{1}{2}mv^2 = eV$$

2 Write the kinetic energy in terms of momentum. Since $p = mv$, $v = p/m$:

$$\tfrac{1}{2}m\left(\frac{p}{m}\right)^2 = \frac{p^2}{2m} = eV$$

3 Solve for the momentum:

$$p = \sqrt{2meV}$$

4 Put this into the de Broglie equation:

$$\boxed{\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}}}$$

So $\lambda \propto 1/\sqrt{V}$: four times the voltage halves the wavelength, and roughly halves the ring sizes.

WORKED EXAMPLE

Electrons are accelerated from rest through a p.d. of 4.0 kV. Calculate their de Broglie wavelength, and explain why a tennis ball of mass 0.058 kg moving at 30 m s⁻¹ shows no diffraction effects.

Show solution

Momentum of the electron from $p = \sqrt{2meV}$:

$$p = \sqrt{2 \times 9.11 \times 10^{-31} \times 1.60 \times 10^{-19} \times 4000} = 3.42 \times 10^{-23}\ \text{kg m s}^{-1}$$

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{3.42 \times 10^{-23}} = 1.94 \times 10^{-11}\ \text{m}$$

This is comparable to the spacing of atoms in a crystal, so the electrons diffract noticeably. For the tennis ball:

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{0.058 \times 30} = 3.8 \times 10^{-34}\ \text{m}$$

That is far smaller even than a proton (about 10^{-15} m across). There is no gap anywhere near that small, so the ball's diffraction is far too small ever to detect.

MARK-SCHEME WORDING

- "Use of p = mv", "use of λ = h/p"; with a p.d., "use of eV = ½mv²".
- Reverse questions: find m from λ and v, then say what the particle is (9.1 × 10⁻³¹ kg → electron; "similar" accepted, WPH12 Jan 2023).
- Some parts require "at least 3 s.f." in a show-that (WPH12 Jan 2026).

Often asked as: direct λ calculations, electrons through a p.d., rearrangement MCQs (v = h/mλ), a car or person's wavelength, find the particle from its mass.

COMMON MISTAKE

Dividing h by the **velocity** instead of the momentum (scored 0, WPH12 Jan 2025 examiner report).

Putting the electron **charge** in where the **mass** belongs.

Leaving off the conclusion when the question ends with a claim to check.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Momentum straight from kinetic energy: $E_k = p^2/2m$, so $p = \sqrt{(2mE_k)}$ (accepted, WPH12 Jan 2026).
- **Photons have momentum** too: $p = h/\lambda$ (WPH12 Jan 2021, MCQ).

SPEC 57 The wave model and the photon model of light

WHAT THIS SPEC POINT ASKS

Understand how electromagnetic radiation can be described by a wave model and by a photon model, and how these models developed over time.

Two models, each needed

Behaviour of light	Explained by
Reflection, refraction	both models (the wave model correctly predicts light slows in glass)
Diffraction, interference, polarisation	**wave model** only
Photoelectric effect (59–62), line spectra (63)	**photon model** only

- **Wave model:** light is a transverse electromagnetic wave. Its energy is spread continuously over the wavefront, and the energy delivered depends on **intensity** (amplitude squared) and time.
- **Photon model:** light travels as **photons**, discrete packets (quanta) of energy $E = hf$. Each photon interacts with **one** electron and gives up **all** its energy at once.
- Light needs both models: this is **wave–particle duality**. Electrons show it too (53).

How the ideas developed

When	Who	Step
late 1600s	Newton	light as a stream of **particles (corpuscles)**; he wrongly predicted light speeds up in glass
late 1600s	Huygens	light as a **wave**, using secondary wavelets

When	Who	Step
1801	Young	**two-slit interference**: strong evidence for waves; wave model accepted
1850	Foucault	light measured to be **slower** in water: agrees with waves, not Newton
1860s–1880s	Maxwell, then Hertz	light is an **electromagnetic wave**; Hertz makes radio waves and notices that UV helps sparks jump (the photoelectric effect)
1900	Planck	energy is emitted in **quanta**, E = hf
1905	Einstein	light itself comes in **photons**: explains the photoelectric effect
1916	Millikan	experiments confirm Einstein's photoelectric equation
1924–1927	de Broglie; Davisson and Germer, G. P. Thomson	particles have wavelengths; **electron diffraction** confirms it

A model is kept while it explains the evidence. When a new observation (like the photoelectric effect) can't be explained, the model is extended or a new one is added.

MARK-SCHEME WORDING

The marks are for **what each model predicts**, not dates:

- Wave model: energy builds up over time, so emission would be **delayed** in dim light; **any frequency** would work given enough intensity or time; electron energy would depend on **intensity**.
- Photon model: a **minimum (threshold) frequency** is needed; emission is **instantaneous**; kinetic energy depends on **frequency**; intensity changes the **number** of photons, not their energy.

Often asked as: MCQ on which model explains which effect; 6-mark comparisons of the two models using the photoelectric effect (see 62).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

"In the wave model, more intense light means a bigger amplitude, so any intensity would eventually release electrons. In the photon model, intensity changes the number of photons, not their energy, so only UV photons with enough energy release electrons" (5 marks, WPH12 Jan 2025).

SPEC 58 Photon energy, E = hf

WHAT THIS SPEC POINT ASKS

Use E = hf to link the energy of a photon to the frequency of the radiation.

The equation

A **photon** is a discrete packet (quantum) of electromagnetic energy. It has no mass and no charge.

Photon energy

FORMULA SHEET

$$E = hf$$

E: energy of one photon / J $h = 6.63 \times 10^{-34}$ J s f: frequency / Hz

Using $c = f\lambda$

$$E = \frac{hc}{\lambda}$$

λ: wavelength / m $c = 3.00 \times 10^{8}$ m s^{-1}

E = photon energy (J), h = 6.63 × 10⁻³⁴ J s, f = frequency (Hz), c = 3.00 × 10⁸ m s⁻¹, λ = wavelength (m). E = hf is on the formulae sheet.

- Higher frequency (shorter wavelength) → more energy per photon. Gamma and X-ray photons carry the most; radio photons the least.
- **Photons per second** from a source of power P: N = P / E = P / hf. For a beam, P = intensity × area (44).
- For the same power, shorter-wavelength light means fewer photons per second, because each carries more energy.

WORKED EXAMPLE

photon energy and photon rate

(a) Blue light, λ = 450 nm: E = hc/λ = (6.63 × 10⁻³⁴ × 3.00 × 10⁸) ÷ (450 × 10⁻⁹) = **4.42 × 10⁻¹⁹ J** (3 s.f.) = 2.76 eV.

(b) A 5.0 mW laser emits 633 nm light. E per photon = 1.99 × 10⁻²⁵ ÷ 633 × 10⁻⁹ = 3.14 × 10⁻¹⁹ J.

Photons per second = P/E = 5.0 × 10⁻³ ÷ 3.14 × 10⁻¹⁹ = **1.6 × 10¹⁶ s⁻¹** (2 s.f.).

WORKED EXAMPLE

A laser pointer emits red light of wavelength 633 nm with a power of 1.0 mW. Calculate (a) the energy of one photon and (b) the number of photons emitted per second.

Show solution

(a) Use $E = hc/\lambda$, with the wavelength in metres:

$$E = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{633 \times 10^{-9}} = 3.14 \times 10^{-19} \text{ J}$$

(b) Power is energy per second, so divide it by the energy of one photon:

$$N = \frac{P}{E} = \frac{1.0 \times 10^{-3}}{3.14 \times 10^{-19}} = 3.2 \times 10^{15} \text{ photons per second}$$

MARK-SCHEME WORDING

- "Use of v = fλ" (or c = fλ) and "use of E = hf". A single combined substitution into E = hc/λ earns both marks.
- Photon counting: "E = Pt", then "number = total energy ÷ energy of one photon".

Often asked as: on nearly every paper: E from λ (and back), photons emitted per second or per minute, percentage of photons that release an electron.

COMMON MISTAKE

Forgetting nm → m (× 10^{-9}).

In a "show that", giving only the rounded value from the question. Show more figures than it gives (e.g. 4.43 eV for "4.4 eV").

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Photocell efficiency: electrons per second = current ÷ e (I/e), compared with photons per second arriving, e.g. 82 % of photons release an electron (WPH12 Jan 2026).

SPEC 59 Absorbing a photon can release a photoelectron

WHAT THIS SPEC POINT ASKS

Understand that when a photon is absorbed, an electron (a photoelectron) can be emitted.

The photoelectric effect

When electromagnetic radiation of high enough frequency shines on a metal surface, **electrons are emitted** from the surface. These are called **photoelectrons**.

What happens, step by step

1. A **photon** hits the metal and is **absorbed by one electron**: a **one-to-one** interaction. The photon gives up **all** its energy; it can't be partly absorbed.
2. Part of that energy is needed to free the electron from the metal (at least the **work function**, 60).
3. If the photon's energy is enough, the electron escapes. Whatever energy is left over becomes its **kinetic energy**.
4. Electrons from deeper inside lose some energy on the way out, so photoelectrons leave with a **range** of kinetic energies, up to a **maximum**.
5. If the photon doesn't have enough energy, no electron escapes, however many photons arrive. An electron can't save up energy from two photons.

Intensity and frequency

- **Brighter light** (same frequency) = **more photons per second** → more electrons emitted per second → bigger photocurrent. The maximum kinetic energy is unchanged.
- **Higher frequency** = more energy per photon → **faster** photoelectrons (more kinetic energy). The number per second depends on how many photons arrive.

The gold-leaf electroscope demonstration

- A clean **zinc** plate on a **negatively** charged electroscope: the leaf stands up.
- Shine **UV** on it: the leaf **falls** at once. Photoelectrons leave, so the plate loses its negative charge.
- Shine bright **visible** light: nothing happens. Its photons have too little energy for zinc.
- Charge the plate **positively**: UV makes no difference, because any released electrons are pulled back.

MARK-SCHEME WORDING

- "A photon is absorbed by an electron"; "**one photon interacts with one electron**"; "the electron gains the photon's energy and is emitted / escapes".
- "Energy is shared between the work function and the kinetic energy of the electron".
- Intensity: "more photons **per second**, so more electrons **per second**, so a greater current". The idea of a rate is needed.

Often asked as: describe the process (2–4 marks), why photoelectrons have a range of speeds, the effect of intensity on the current, MCQs (a photon can't be partly absorbed).

COMMON MISTAKE

Writing answers where it isn't clear whether photons or electrons are being released (WPH12 Oct 2020 examiner report). Photons go **in**; electrons come **out**.

Saying brighter light gives faster electrons.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

A photocell has a metal cathode and a collecting gauze. With the gauze positive, electrons are attracted across and a current flows; reverse the polarity and the current stops once the p.d. is large enough (WPH12 Oct 2023).

SPEC 60 Threshold frequency, work function and hf = φ + ½mv²max

WHAT THIS SPEC POINT ASKS

Know what threshold frequency and work function mean, and use hf = φ + ½mv²max.

Definitions

- **Work function φ**: the **minimum** energy needed to remove an electron from the **surface** of a metal. Each metal has its own value.
- **Threshold frequency f_0**: the **minimum** frequency of radiation that will release electrons from the metal. A photon at f_0 has just enough energy: $\mathbf{hf_0 = \phi}$, so $f_0 = \phi/h$. The matching threshold wavelength is $\lambda_0 = c/f_0$ (longer wavelengths won't work).

Einstein's photoelectric equation

Energy conservation for the most energetic electrons (those from the surface):

Einstein's equation

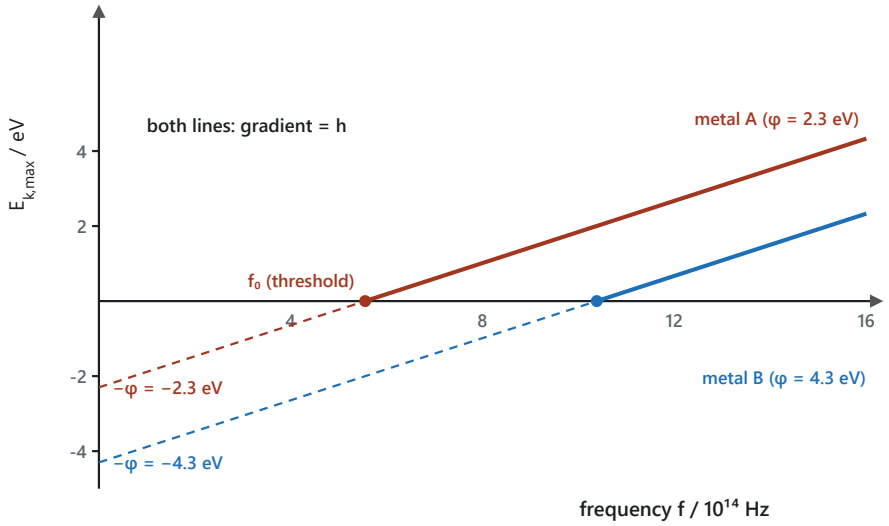
FORMULA SHEET

$$hf = \phi + \tfrac{1}{2}mv^2_{\max}$$

hf: energy of one photon / J ϕ: work function / J $\tfrac{1}{2}mv^2_{\max}$: maximum kinetic energy of a photoelectron / J

hf = photon energy (J), φ = work function (J), ½mv²max = maximum kinetic energy of a photoelectron (J), m = 9.11 × 10⁻³¹ kg. On the formulae sheet.

The graph



E_k,max = hf − φ. Each metal gives a straight line of gradient h. It meets the f-axis at the threshold frequency f₀ and, extended back, the y-axis at −φ. A metal with a bigger φ gives a parallel line further right.

Stopping potential

Make the collector negative and increase the p.d. until the current just stops. Then even the fastest electrons are turned back, so the work done equals their kinetic energy:

Stopping potential

$$eV_s = \tfrac{1}{2}mv^2_{\max}$$

e: electron charge = 1.60 × 10⁻¹⁹ C V_s: stopping potential / V
$\tfrac{1}{2}mv^2_{\max}$: maximum kinetic energy of a photoelectron / J

WORKED EXAMPLE

zinc in ultraviolet

Zinc has φ = 4.3 eV. UV of wavelength 200 nm shines on it.

Photon energy: $E = hc/\lambda = 1.99 \times 10^{-25} \div 200 \times 10^{-9} = 9.95 \times 10^{-19}$ J.

φ in joules: $4.3 \times 1.60 \times 10^{-19} = 6.88 \times 10^{-19}$ J.

$E_{k,max} = hf - \varphi = 9.95 \times 10^{-19} - 6.88 \times 10^{-19} = 3.07 \times 10^{-19}$ J.

$v_{max} = \sqrt{(2E_k/m)} = \sqrt{(2 \times 3.07 \times 10^{-19} \div 9.11 \times 10^{-31})} =$ **8.2×10^5 m s^{-1}** (2 s.f.).

Stopping potential: $V_s = E_{k,max}/e = 3.07 \times 10^{-19} \div 1.60 \times 10^{-19} =$ **1.9 V**.

Threshold frequency: $f_0 = \varphi/h = 6.88 \times 10^{-19} \div 6.63 \times 10^{-34} =$ **1.0×10^{15} Hz** (threshold wavelength 290 nm). Visible light (400–700 nm) can't release electrons from zinc.

Where Einstein's photoelectric equation comes from

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. One photon is absorbed by one electron, so the energy the electron receives is the whole photon energy:

$$E_{\text{in}} = hf$$

2. To leave the metal, the electron must do work against the attraction of the positive ions. The least work any electron has to do is the work function ϕ, for an electron right at the surface.

$$W_{\text{escape}} \geq \phi$$

3. Energy is conserved, so whatever is left over becomes kinetic energy:

$$\tfrac{1}{2}mv^2 = hf - W_{\text{escape}} \leq hf - \phi$$

This is why the electrons come out with a spread of speeds: different electrons lose different amounts of energy on the way out.

4. The fastest electrons are the ones that only had to do work ϕ. Rearranging:

$$\boxed{hf = \phi + \tfrac{1}{2}mv_{\text{max}}^2}$$

5. At the threshold frequency the photon has only just enough energy, so the maximum kinetic energy is zero:

$$hf_0 = \phi + 0 \quad \Rightarrow \quad \boxed{f_0 = \frac{\phi}{h}}$$

Below f_0, $hf - \phi$ would be negative: no electron can escape, however many photons arrive.

WORKED EXAMPLE

Ultraviolet radiation of wavelength 250 nm falls on a zinc plate. The work function of zinc is 4.3 eV. Calculate (a) the maximum kinetic energy of the photoelectrons in J, (b) their maximum speed and (c) the threshold frequency for zinc.

Show solution

(a) Photon energy, and the work function in joules:

$$hf = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{250 \times 10^{-9}} = 7.96 \times 10^{-19}\ \text{J}$$

$$\phi = 4.3 \times 1.60 \times 10^{-19} = 6.88 \times 10^{-19}\ \text{J}$$

$$\tfrac{1}{2}mv_{\max}^2 = hf - \phi = 7.96 \times 10^{-19} - 6.88 \times 10^{-19} = 1.08 \times 10^{-19}\ \text{J}$$

(b) Rearrange the kinetic energy for v, using $m_e = 9.11 \times 10^{-31}$ kg:

$$v_{\max} = \sqrt{\frac{2 \times 1.08 \times 10^{-19}}{9.11 \times 10^{-31}}} = 4.9 \times 10^{5}\ \text{m s}^{-1}$$

(c) At threshold, $hf_0 = \phi$:

$$f_0 = \frac{\phi}{h} = \frac{6.88 \times 10^{-19}}{6.63 \times 10^{-34}} = 1.0 \times 10^{15}\ \text{Hz}$$

That's a wavelength of $c/f_0 \approx 290$ nm. Visible light (above 380 nm) can never release electrons from zinc.

MARK-SCHEME WORDING

- Work function: "**minimum** energy required to release/remove an electron from the **surface** of a metal". "Energy needed" without "minimum" is ignored.
- "Converts φ from eV to J" is a marking point. "Use of hf = φ + ½mv²max."
- The speed mark needs real energy and mass values substituted (not the charge).
- Graph: "gradient = h", "x-intercept = threshold frequency", "y-intercept = −φ".

Often asked as: define φ, find v_max, identify a metal from φ in a table, decide whether emission happens, threshold frequency, stopping potential, read h and φ from a graph.

COMMON MISTAKE

Forgetting "minimum" in the work function definition (WPH12 Oct 2020 examiner report).

Mixing eV and J in one subtraction.

Saying brighter light increases the electrons' speed. Only frequency does (WPH12 Jan 2021).

No unit on v_max or V_s.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- A photoelectron then **accelerated** through a p.d. V gains eV, so final E_k = initial E_k + eV, e.g. 8.30 × 10⁻¹⁹ J + 1.73 × 10⁻¹⁷ J = 1.81 × 10⁻¹⁷ J (WPH12 Oct 2024).
- One paper (WPH12 Jun 2022) gave a graph with deliberately wrong intercepts and asked you to show it can't be zinc. Don't learn its −10 eV intercept as data.

SPEC 61 The electronvolt

WHAT THIS SPEC POINT ASKS

Use the electronvolt (eV) to express small energies.

Definition

1 electronvolt is the energy transferred when **one electron** moves through a **potential difference of 1 volt**. From W = QV:

The electronvolt

$$1\ \text{eV} = 1.60 \times 10^{-19}\ \text{C} \times 1\ \text{V} = 1.60 \times 10^{-19}\ \text{J}$$

- **eV → J:** multiply by 1.60 × 10⁻¹⁹. **J → eV:** divide by 1.60 × 10⁻¹⁹.
- An electron accelerated from rest through V volts gains **V electronvolts** of kinetic energy.
- Photon energies, work functions and atomic energy levels are a few eV, so the eV keeps the numbers simple.

WORKED EXAMPLE

conversions

(a) An electron accelerated from rest through 150 V gains 150 eV = 150 × 1.60 × 10⁻¹⁹ = 2.40 × 10⁻¹⁷ J. Its speed: v = √(2 × 2.40 × 10⁻¹⁷ ÷ 9.11 × 10⁻³¹) = **7.26 × 10⁶ m s⁻¹** (3 s.f.).

(b) A 700 nm photon: E = 1.99 × 10⁻²⁵ ÷ 700 × 10⁻⁹ = 2.84 × 10⁻¹⁹ J = 2.84 × 10⁻¹⁹ ÷ 1.60 × 10⁻¹⁹ = **1.78 eV**.

MARK-SCHEME WORDING

- "Converts J to eV" / "eV to J" is usually its own mark.
- In a comparison either unit is fine if both values are in the same unit (e.g. 0.66 eV < 1.5 eV, WPH12 Jan 2025).

Often asked as: a one-mark conversion inside photoelectric and energy-level questions.

COMMON MISTAKE

Treating the eV as a unit of voltage. It is a unit of **energy**.

Comparing a value in eV with one in J.

SPEC 62 The photoelectric effect as evidence for the particle nature of light

WHAT THIS SPEC POINT ASKS

Understand how the photoelectric effect gives evidence that electromagnetic radiation behaves as particles.

Observation, wave prediction, photon explanation

Observation	What the wave model predicts	Photon explanation
No electrons below a **threshold frequency**, however bright the light	any frequency should work if the light is intense enough (or shone for long enough)	each photon has energy hf; below f_0 one photon has less than φ, and one electron can't combine energy from several photons
Emission is **instantaneous**, even in very dim light	in dim light an electron would have to absorb energy for a long time before it could escape	one photon gives all its energy to one electron at once
Maximum kinetic energy depends on frequency, not intensity	more intense light (bigger amplitude) should give faster electrons	$E_{k,max} = hf - φ$: only frequency changes the photon energy
Number of electrons per second depends on **intensity**	(not a clear prediction)	more intense = more photons per second = more electrons per second

Writing the 6-mark answer

1. Light arrives as **photons**, each with energy **E = hf**.
2. **One photon** interacts with **one electron**.
3. An electron needs a **minimum energy (the work function)** to escape, so there's a **threshold frequency**.
4. So emission is **instantaneous** when $f > f_0$; in the wave model energy would have to build up.
5. Extra energy becomes kinetic energy, so **frequency** controls $E_{k,max}$; **intensity** only controls the **number** of electrons per second.
6. These can only be explained if light behaves as **particles**.

WORKED EXAMPLE · GRAPH

A student measures the maximum kinetic energy of photoelectrons from a metal for several frequencies and draws a best-fit line. Two points on the line are $(7.0 \times 10^{14}\ \text{Hz},\ 0.62\ \text{eV})$ and $(1.10 \times 10^{15}\ \text{Hz},\ 2.28\ \text{eV})$. (a) Use the graph to find a value for the Planck constant. (b) Find the work function of the metal in eV. (c) Calculate the threshold frequency.

Show solution

(a) The gradient of the KE–f graph is h. Work it out in eV s, then convert to J s:

$$\text{gradient} = \frac{2.28 - 0.62}{1.10 \times 10^{15} - 7.0 \times 10^{14}} = \frac{1.66\ \text{eV}}{4.0 \times 10^{14}\ \text{Hz}} = 4.15 \times 10^{-15}\ \text{eV s}$$

$$h = 4.15 \times 10^{-15} \times 1.60 \times 10^{-19} = 6.64 \times 10^{-34}\ \text{J s}$$

(b) Rearrange $\frac{1}{2}mv_{\max}^2 = hf - \phi$ and use either point:

$$\phi = hf - \tfrac{1}{2}mv_{\max}^2 = (4.15 \times 10^{-15} \times 7.0 \times 10^{14}) - 0.62 = 2.91 - 0.62 = 2.29\ \text{eV} \approx 2.3\ \text{eV}$$

(c) At threshold, $f_0 = \phi/h$:

$$f_0 = \frac{2.29 \text{ eV}}{4.15 \times 10^{-15} \text{ eV s}} = 5.5 \times 10^{14} \text{ Hz}$$

This is where the line would cross the frequency axis. The metal is probably sodium.

MARK-SCHEME WORDING

Indicative content credited (WPH12 Jan 2022): photon energy related to frequency; one-to-one interaction; minimum energy/frequency needed; instantaneous release; frequency affects kinetic energy; intensity affects the number of electrons per second. Each "observation" needs its "explanation" for the linkage marks.

Often asked as: 6-mark questions (WPH12 Jun 2019, Jan 2022, Oct 2023), 5-mark wave v photon comparisons, why UV releases electrons but visible light doesn't, two sources and two metals.

COMMON MISTAKE

Saying "brighter light gives more energetic electrons".

Describing the observations without saying why the **wave** model fails, when the question asks you to compare.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

A wave-model estimate of the delay: a single atom (area about 8×10^{-20} m²) in dim light would take about two minutes (118 s) to collect enough energy, yet emission is instant (WPH12 Oct 2020).

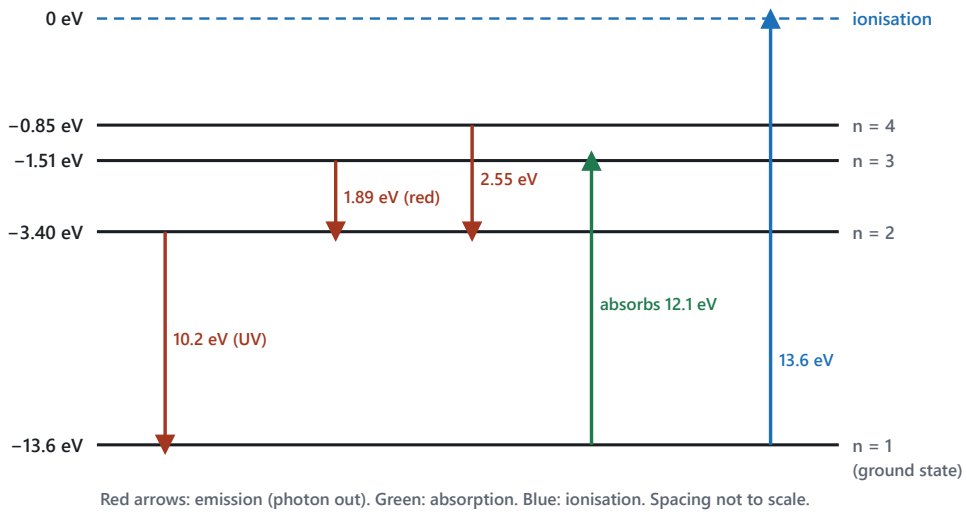
SPEC 63 Line spectra and energy levels

WHAT THIS SPEC POINT ASKS

Explain atomic line spectra using transitions between discrete energy levels, and calculate the frequency of the radiation emitted or absorbed in a transition.

Discrete energy levels

- Electrons in an atom can only have certain **discrete (fixed) energies**: the **energy levels**. Each element has its own set.
- The levels are **negative**: 0 eV is the energy of an electron that has just been removed (**ionisation**). The lowest level is the **ground state**.
- **Ionisation energy** = the energy needed to remove an electron from the ground state completely (13.6 eV for hydrogen).



Hydrogen. Emission arrows point down (photon out), absorption arrows point up (photon in). The bigger the jump, the bigger the photon energy and the shorter the wavelength.

Emission spectra

1. Atoms gain energy (in a discharge tube, collisions with fast electrons; or by heating).
2. An electron is **excited** to a higher energy level.
3. It is unstable there and drops back to a lower level, **emitting a photon** whose energy equals the **difference** between the two levels.
4. The levels are discrete, so only certain energy differences exist, so only **certain frequencies (wavelengths)** are emitted: a **line spectrum** of bright lines on a dark background.

Photon from a transition

$$hf = E_1 - E_2$$

E_1: energy of the higher level / J E_2: energy of the lower level / J f: frequency of the photon / Hz

E_1 = upper level, E_2 = lower level (J; convert from eV). f = frequency of the photon (Hz).

Absorption spectra

- Shine white light (all wavelengths) through a cool gas. Photons with **exactly** the energy of an upward transition are absorbed; all others pass straight through.
- You see a continuous spectrum crossed by **dark lines** at the **same** wavelengths as that gas's emission lines.
- A photon with the wrong energy is not absorbed at all, even if it has more than enough. (An electron colliding with the atom is different: it can hand over just part of its kinetic energy.)

WORKED EXAMPLE

hydrogen transitions

(a) n = 3 (−1.51 eV) to n = 2 (−3.40 eV): ΔE = 3.40 − 1.51 = 1.89 eV = 1.89 × 1.60 × 10^{-19} = 3.02 × 10^{-19} J.

f = ΔE/h = 3.02 × 10⁻¹⁹ ÷ 6.63 × 10⁻³⁴ = **4.56 × 10¹⁴ Hz**; λ = c/f = **6.58 × 10⁻⁷ m (658 nm, red)**.

(b) n = 2 to n = 1: ΔE = 13.6 − 3.40 = 10.2 eV = 1.63 × 10⁻¹⁸ J, f = 2.46 × 10¹⁵ Hz, λ = **1.22 × 10⁻⁷ m** (ultraviolet).

(c) Can a 12.3 eV photon be absorbed by hydrogen in the ground state? −13.6 + 12.3 = −1.3 eV, which is not a level (they are −1.51 and −0.85). So **no**: the photon passes through and the electron stays in the ground state.

How many lines?

An electron in level n can come down in steps or in one jump. From n = 4 the possible jumps are 4→3, 4→2, 4→1, 3→2, 3→1, 2→1: **six** lines.

WORKED EXAMPLE

An electron in a hydrogen atom drops from the level at $-1.51\ \text{eV}$ to the level at $-3.40\ \text{eV}$. Calculate the wavelength of the photon emitted and state the region of the EM spectrum it is in.

Show solution

Energy difference, converted to joules:

$$\Delta E = -1.51 - (-3.40) = 1.89\ \text{eV} = 1.89 \times 1.60 \times 10^{-19} = 3.02 \times 10^{-19}\ \text{J}$$

Frequency from $hf = \Delta E$, then wavelength from $c = f\lambda$:

$$f = \frac{3.02 \times 10^{-19}}{6.63 \times 10^{-34}} = 4.56 \times 10^{14}\ \text{Hz} \qquad \lambda = \frac{3.00 \times 10^{8}}{4.56 \times 10^{14}} = 6.58 \times 10^{-7}\ \text{m}$$

658 nm is **visible red light**. (The measured line is at 656 nm; the small difference comes from rounding the level energies to 3 significant figures.)

WORKED EXAMPLE · ENERGY LEVELS

Four of the energy levels of a mercury atom are $-10.4\ \text{eV}$ (ground state), $-5.5\ \text{eV}$, $-3.7\ \text{eV}$ and $-1.6\ \text{eV}$. (a) Calculate the wavelength of the photon emitted in the transition from $-5.5\ \text{eV}$ to the ground state, and name the region of the spectrum. (b) How many different photon energies could be emitted by atoms excited to the $-1.6\ \text{eV}$ level? (c) What is the ionisation energy of mercury in joules?

Show solution

(a) Energy gap in joules, then $\lambda = hc/\Delta E$:

$$\Delta E = -5.5 - (-10.4) = 4.9\ \text{eV} = 4.9 \times 1.60 \times 10^{-19} = 7.84 \times 10^{-19}\ \text{J}$$

$$\lambda = \frac{hc}{\Delta E} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{7.84 \times 10^{-19}} = 2.54 \times 10^{-7}\ \text{m} = 254\ \text{nm}$$

This is **ultraviolet** (shorter than 380 nm). Fluorescent lamps use exactly this line: a coating on the inside of the tube absorbs the UV and emits visible light.

(b) There are four levels, so the number of possible downward jumps is $3 + 2 + 1 = 6$ different photon energies.

(c) Ionisation means going from the ground state to 0 eV: $10.4\ \text{eV} = 10.4 \times 1.60 \times 10^{-19} = 1.66 \times 10^{-18}\ \text{J}$.

MARK-SCHEME WORDING

- "**Electrons** gain energy / are excited", "move to **higher** energy levels", "fall to **lower** levels", "**emitting photons**", "energy levels are **discrete**", "photon energy = **difference** between levels", "E = hf so only certain frequencies".
- Not "atoms" for electrons in some schemes (WPH12 Jun 2019); "gain energy" alone wasn't enough for the first point in another (WPH12 Oct 2021).
- Absorption: "a photon can only excite an electron if its energy **matches** the difference between levels".

Often asked as: 6-mark explanations of a hydrogen or discharge-tube spectrum, λ or f from a transition, which transition gives a given photon, is a given photon absorbed, MCQs on allowed jumps and highest frequency.

COMMON MISTAKE

Thinking a bigger jump gives a longer wavelength. Bigger jump → bigger energy → **shorter** wavelength (WPH12 Oct 2022 examiner report).

Assuming an electron in n = 3 can only drop to n = 2 or n = 1 in one go. Count every possible jump, including n = 3 → 2 → 1 (WPH12 Jan 2023 examiner report).

Talking about the work function in an energy-level question. That belongs to metals and the photoelectric effect (WPH12 Jan 2021 examiner report).

Quoting a transition the wrong way round: emission goes **down**, absorption goes **up**.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Air glows in several colours because it contains different gases, each with its own energy levels (WPH12 Oct 2021).
- Mercury lamps: raising the p.d. doesn't change the UV frequency because the energy levels are fixed (WPH12 Jun 2025).
- Equations for the Bohr atom may be given in the stem; just substitute (e.g. ground-state energy 13.56 eV, WPH12 Oct 2020).

Quick checks

QUICK CHECK

Light above the threshold frequency shines on a metal. The intensity is doubled but the frequency stays the same. What happens?

- The maximum kinetic energy of the photoelectrons doubles.
- Twice as many photoelectrons are emitted per second, with the same maximum kinetic energy.
- The threshold frequency halves.
- The photoelectrons are emitted after a shorter delay.

Doubling the intensity doubles the number of photons per second. Each photon still has the same energy hf, so each electron gets the same energy: more electrons, same $\frac{1}{2}mv_{\max}^2 = hf - \phi$. There is no delay to shorten; emission is already instant.

QUICK CHECK

What is the gradient of a graph of maximum kinetic energy of photoelectrons (in J) against frequency of the radiation?

- The work function ϕ
- The threshold frequency f_0
- The Planck constant h
- The electron charge e

$\frac{1}{2}mv_{\max}^2 = hf - \phi$ is $y = mx + c$ with gradient h. The y-intercept is $-\phi$ and the x-intercept is f_0. Every metal gives the same gradient.

QUICK CHECK

A photon has an energy of 3.0 eV. What is its energy in joules?

- 1.9×10^{19} J
- 4.8×10^{-19} J
- 5.3×10^{-20} J
- 2.0×10^{-34} J

eV → J means multiply by 1.60×10^{-19}: $3.0 \times 1.60 \times 10^{-19} = 4.8 \times 10^{-19}$ J. The first option is what you get by dividing instead of multiplying.

QUICK CHECK

The accelerating voltage in an electron diffraction tube is increased from 2.0 kV to 8.0 kV. By what factor does the de Broglie wavelength of the electrons change?

- It becomes 4 times longer.
- It becomes 4 times shorter.
- It becomes 2 times longer.
- It becomes 2 times shorter.

$\lambda = h/\sqrt{2meV}$, so $\lambda \propto 1/\sqrt{V}$. Four times the voltage gives $\sqrt{4} = 2$ times the momentum, so half the wavelength, and the rings shrink.

QUICK CHECK

The ground state of hydrogen is -13.6 eV and the next levels are -3.40, -1.51 and -0.85 eV. Which photon could be absorbed by a hydrogen atom in its ground state?

- 5.1 eV
- 10.2 eV
- 11.0 eV
- 12.5 eV

A photon is only absorbed if its energy exactly matches a gap: $-3.40 - (-13.6) = 10.2$ eV. The other gaps from the ground state are 12.1 eV and 12.75 eV, and none of the other options matches any of them (and none is enough to ionise the atom).

QUICK CHECK

Which observation can the wave model of light *not* explain?

- Light spreads out after passing through a narrow slit.
- Light can be polarised.
- Red light never releases electrons from zinc, however bright it is.
- More photoelectrons are emitted when the light is brighter.

In the wave model, brighter light delivers more energy, so eventually any frequency should release electrons. The threshold frequency only makes sense if energy comes in photons of hf, one per electron. Diffraction and polarisation are wave behaviour, and the number of electrons rising with brightness fits both models.